

Beyond Visual Perception: Robust Chinese Text Recognition via Distilled Context Injection

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A Additional Experimental Results

A.1 Experimental Results for Conventional OCR methods

In this section, we evaluate the effectiveness of the proposed CCTR on page-level and line-level text recognition tasks. Specifically, we report the results under both AR(AR*) and CR(CR*) metrics across sixteen subsets from CLTW and HW, aiming to comprehensively assess the model’s robustness under diverse challenging conditions, including degradation, occlusion, and distortion. The detailed comparisons with existing OCR methods are presented as follows:

Tab. 1 presents AR* as the evaluation metric on sixteen subsets from CLTW and HW, comparing CCTR with multiple known OCR models. The results show that: 1) CCTR achieves the best overall performance with an average accuracy of 90.98%, outperforming SVTRv2 by 0.43 percentage points. Although the improvement appears small, it is notable under such a saturated metric, where most methods already perform strongly. This gain mainly comes from incorporating a large language model, whose contextual knowledge is distilled to enhance semantic understanding. 2) CCTR consistently outperforms or matches the best results across most subsets. The largest improvements are observed on challenging degraded subsets, such as SSB Pair (Degr) (+0.59%) and OCC Pair (Degr) (+0.72%), demonstrating stronger robustness to occlusion and degradation. 3) SMTR outperforms CCTR in PER(Degr), where semantic information is heavily degraded, limiting the effectiveness of semantic priors and making accurate visual modeling more critical.

Tab. 2 summarizes the CR* performance on sixteen subsets from CLTW and HW, with comparisons between CCTR and a range of existing OCR methods. The results demonstrate that: 1) CCTR achieves the best overall performance with 91.76%, surpassing SVTRv2 by 0.43 percentage points. Similar to AR*, this improvement is significant given the already high performance level, indicating the effectiveness of our method at character-level recognition. 2) SMTR outperforms our CCTR method on CR* across several subsets, yet underperforms it on CR as shown in Tab. 3. As defined in Eq. ??, the CR metric does not account for insertion errors. As a result, repetitive predictions have limited impact on CR but significantly degrade AR. It tends to produce locally correct but repetitive predictions, which inflate CR scores. Moreover, page-level evaluation is less sensitive to accumulated errors than line-level, leading to generally higher scores for such methods. 3) A larger gap is observed on PER(Degr), where CCTR underperforms SMTR. This is likely because such heavily degraded samples rely more on fine-grained visual modeling, while our method focuses more on high-level semantic enhancement via language model distillation.

Tab. 3 and Tab. 4 report the line-level recognition performance on CLTW and HW benchmarks using two complementary metrics: CR and AR, respectively. we have the following observations: 1) CCTR achieves the best performance on AR (91.64%) and CR (92.18%), consistently outperforming

Table 1: Page-level recognition accuracy (AR*) on CLTW and HW.

Method	UND Pair		REF Pair		SSB Pair		PER Pair		OCC Pair		SCR	CBG	SPM	HW	OCF	HAL	Avg
	Norm	Degr	Norm	Degr	Norm	Degr	Norm	Degr	Norm	Degr							
SAR	52.11	51.98	56.89	51.86	55.05	54.07	62.14	39.48	50.25	41.78	41.70	41.91	51.13	72.10	47.91	45.89	51.02
ABInet	58.56	57.77	61.38	53.68	61.41	56.07	67.78	43.30	56.17	45.00	49.23	47.74	57.51	71.97	52.28	48.68	55.53
MAERec	62.95	61.84	66.42	58.26	66.59	58.76	72.49	49.97	62.21	50.92	54.63	53.22	60.91	81.89	57.14	52.92	60.70
CRNN	92.55	93.76	88.22	76.43	93.95	82.47	92.93	42.36	85.42	70.69	92.80	89.06	81.32	76.09	80.70	66.20	81.56
LISTER	90.05	90.69	86.52	77.42	92.35	84.53	91.96	43.07	85.55	70.20	89.54	87.98	84.44	80.53	78.68	66.42	81.25
IGTR_AR	63.59	63.93	68.39	63.42	67.04	64.84	72.75	43.47	64.35	55.32	56.60	55.72	67.32	80.88	57.90	54.77	64.35
SMTR	74.08	68.44	77.30	74.72	74.59	73.60	76.02	70.54	67.87	54.43	61.66	64.89	77.07	72.91	65.65	65.62	69.96
SVTRv2	96.17	96.96	94.11	88.37	97.32	91.86	97.00	65.80	94.05	85.05	95.90	95.66	91.34	90.17	88.65	80.37	90.55
CCTR (ours)	96.45	97.18	94.44	89.05	97.49	92.45	97.20	66.71	94.44	85.77	96.27	96.04	91.54	90.71	89.10	80.90	90.98

Table 2: Page-level recognition accuracy (CR*) on CLTW and HW.

Method	UND Pair		REF Pair		SSB Pair		PER Pair		OCC Pair		SCR	CBG	SPM	HW	OCF	HAL	Avg
	Norm	Degr	Norm	Degr	Norm	Degr	Norm	Degr	Norm	Degr							
SAR	52.38	52.23	57.31	52.41	55.40	54.50	62.48	40.13	50.78	42.40	41.98	42.11	51.54	74.17	48.23	46.39	51.53
ABInet	58.76	58.00	61.94	54.32	61.77	56.45	68.12	43.74	56.61	45.53	49.48	48.01	57.92	74.06	52.55	49.17	56.03
MAERec	63.10	62.00	66.71	58.74	66.84	59.13	72.70	50.43	62.60	51.24	54.95	53.47	61.21	83.63	57.4	53.37	61.10
CRNN	92.65	93.89	88.41	76.68	94.13	82.73	93.05	42.43	85.65	70.86	93.22	89.20	81.47	77.26	80.81	66.39	81.80
LISTER	92.59	93.64	89.73	79.79	95.19	86.67	94.06	47.96	87.54	74.09	93.17	90.89	85.51	85.21	82.05	69.95	84.25
IGTR_AR	63.67	64.02	68.65	63.77	67.28	65.09	72.91	43.75	64.56	55.67	56.81	55.86	67.53	82.58	58.07	55.07	62.83
SMTR	92.13	91.48	92.43	90.25	94.13	91.72	91.42	92.21	88.05	75.47	81.66	91.27	90.12	91.59	89.92	87.81	89.48
SVTRv2	96.35	97.18	94.36	88.76	97.57	92.20	97.19	66.10	94.28	92.20	96.30	95.85	91.61	91.75	88.89	80.73	91.33
CCTR (ours)	96.63	97.40	94.69	89.43	97.74	92.79	97.39	67.01	94.67	92.79	96.67	96.22	91.91	92.29	89.33	81.26	91.76

conventional OCR methods such as SVTRv2, thereby demonstrating the overall effectiveness of the proposed approach. 2) While SMTR achieves competitive or even higher CR scores (88.66% in Tab. 3), its AR performance is noticeably lower (67.11% in Tab. 4). In contrast, CCTR maintains a smaller gap between CR and AR, suggesting that it generates more structurally consistent predictions with fewer redundant or misaligned characters. 3) It is worth noting that SMTR significantly outperforms CCTR on the PER(Degr) subset. We observe that SMTR tends to produce redundant or repeated characters under severe degradation, which boosts CR but leads to misaligned sequences. In contrast, CCTR generates more compact and well-aligned predictions, resulting in more consistent AR performance.

A.2 Examples of Text Recognition Results

To further illustrate the effectiveness of the proposed method, we provide qualitative comparisons.

We compare the recognition results of CCTR with conventional OCR models on challenging samples. As shown in Fig. 1, existing methods often fail under severe degradation, occlusion, or complex layouts, causing missing or incorrect predictions. In contrast, CCTR produces more accurate and complete results. This improvement is attributed to semantic priors distilled from the large language model, enabling better recovery of corrupted or ambiguous characters through contextual information.

Table 3: Line-level recognition accuracy (CR) on CLTW and HW.

Method	UND Pair		REF Pair		SSB Pair		PER Pair		OCC Pair		SCR	CBG	SPM	HW	Avg
	Orig	Degr	Orig	Degr	Orig	Degr	Orig	Degr	Orig	Degr					
SAR	52.20	51.98	57.22	52.19	55.20	54.33	62.37	39.73	50.62	42.46	41.74	41.87	51.32	74.20	51.38
ABInet	58.62	57.84	61.85	54.06	61.62	56.31	68.07	43.31	56.45	45.43	49.33	47.79	57.70	74.10	56.85
PARSeq	80.68	81.06	87.16	78.27	84.24	80.59	86.60	62.45	82.94	75.37	74.92	77.07	82.37	90.46	80.30
MAERec	62.99	61.88	66.65	58.58	66.78	58.99	72.65	50.22	62.56	51.15	54.83	53.34	61.09	83.64	61.57
CRNN	92.63	93.88	88.38	76.56	94.08	82.67	93.03	42.09	85.57	70.74	93.22	89.19	81.41	72.28	82.77
LISTER	92.58	93.66	89.75	79.77	95.16	86.63	94.06	47.98	87.54	74.15	93.20	91.00	85.47	85.25	86.16
IGTR_AR	63.60	63.92	68.62	63.61	67.23	65.00	72.87	43.34	64.45	55.57	56.72	55.79	67.44	82.61	63.91
SMTR	91.65	90.92	92.39	88.91	94.13	90.27	91.50	91.28	87.22	74.10	80.56	91.09	87.14	90.18	88.66
SVTRv2	96.35	97.18	94.35	88.70	97.51	92.19	97.19	65.99	94.29	85.29	96.30	95.86	91.56	91.76	91.89
CCTR(ours)	96.53	97.38	94.66	89.22	97.68	92.61	97.35	66.56	94.62	85.24	96.57	96.16	91.87	92.20	92.18

Table 4: Line-level recognition accuracy (AR) on CLTW and HW.

Method	UND Pair		REF Pair		SSB Pair		PER Pair		OCC Pair		SCR	CBG	SPM	HW	Avg
	Orig	Degr	Orig	Degr	Orig	Degr	Orig	Degr	Orig	Degr					
SAR	51.87	51.67	56.72	51.50	54.75	53.81	61.99	38.68	49.99	41.35	41.36	41.61	50.80	71.86	50.78
ABInet	58.41	57.60	61.27	53.36	61.22	55.88	67.72	42.76	55.99	44.74	49.02	47.51	57.21	71.78	56.03
PARSeq	80.57	80.93	86.74	77.81	83.98	80.29	86.41	62.06	82.59	74.78	74.73	76.83	82.12	88.42	79.91
MAERec	62.84	61.69	66.33	58.05	66.49	58.53	72.43	49.62	62.13	50.60	54.51	53.05	60.74	81.78	61.20
CRNN	92.52	93.75	88.18	76.27	93.86	82.40	92.90	42.00	85.34	70.57	92.80	89.05	81.25	75.94	82.63
LISTER	89.88	90.58	86.02	76.88	92.19	84.30	91.85	39.75	85.45	69.56	89.52	87.92	83.89	80.42	82.02
IGTR_AR	63.52	63.83	68.35	63.23	66.95	64.72	72.69	42.86	64.22	55.10	56.50	55.65	67.22	80.77	63.16
SMTR	70.25	70.39	75.36	68.17	74.40	66.46	76.94	69.97	58.58	51.95	52.52	64.96	69.00	70.63	67.11
SVTRv2	96.17	96.96	94.09	88.29	97.22	91.84	96.99	65.65	94.04	85.01	95.90	95.66	91.30	90.12	91.23
CCTR(ours)	96.35	97.15	94.40	88.81	97.38	92.26	97.16	66.23	94.37	85.57	96.17	96.97	91.60	90.57	91.64



Figure 1: Examples of Recognition Results on Handwriting and CLTW.

Overall, the results further validate that CCTR not only improves quantitative performance but also enhances the model’s robustness and interpretability by introducing explicit semantic guidance during training.

B Symbol Description

Table 5: Summary of notation.

	Symbol	Description
Models	$g_v(\cdot), g_v^*(\cdot)$	VM, Optimal VM
	$g_l(\cdot), g_l^*(\cdot)$	LM, Optimal LM
	$f(\cdot, \cdot), f^*(\cdot, \cdot)$	CMSI, Optimal CMSI
Probability distributions	$\mathbf{p}_v$	Probability distribution of VM
	$\mathbf{p}_l$	Probability distribution of LM
	$\mathbf{p}_f$	Probability distribution of CMSI
Data	$\mathbf{x}$	Vector of input image
	$\mathbf{y}$	Vector of ground-truth label
	$\mathbf{H}$	Input/Output represents of transformer block
	$\mathbf{s}$	Text lines output by vision model
	z_v	Logits of VM
	z_l	Logits of LM
Theory-related	$R(\cdot)$	Population risk
	Q_v, k_v, V_v	Query, key and value of VM
	Q_l, K_l	Query and key of LM
	W_q, W_k, W_v	Learnable cross attention weight in VM to LM direction
	λ	Lipschitz constant
	$\text{Attn}_{\text{pc}}(\cdot, \cdot)$	Cross attention from VM to LM direction
	$\text{Attn}_{\text{ep}}(\cdot, \cdot)$	Cross attention from LM to VM direction
Loss function	$\ell(\cdot)$	Loss function
	L_{KD}	Cross entropy loss of LM
	L_{CE}	Cross entropy loss of CMSI
Others	$\mathcal{D}$	Data distribution
	τ	Temperature coefficient
	$\phi(\cdot)$	Nonlinear mapping of transformer encoder